

Manish Man Shrestha

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PROFILE SUMMARY

Senior Embedded Firmware Engineer with 15 years of total experience, including 8 years of industry experience, developing real-time firmware and low-level software for hardware-centric embedded systems. Strong background in Embedded C/C++, bare-metal and RTOS-based firmware design, microcontroller programming, device drivers, board bring-up, hardware debugging, and firmware-hardware integration. Experienced in optimizing firmware for timing, memory usage, power consumption, reliability, and real-time performance in constrained embedded environments. Hands-on experience with embedded DSP, ROS-based robotic systems, sensor interfaces, PWM/control systems, wireless connectivity, firmware validation, and test/debug tooling using Python and C#/.NET.

CORE SKILLS

- **Embedded Firmware Development:** Embedded C/C++, real-time firmware, device drivers, board bring-up, interrupt-driven systems, low-level hardware programming
- **RTOS & Real-Time Systems:** FreeRTOS, task scheduling, inter-task communication, semaphores, mutexes, queues, timing optimization
- **MCUs / SoCs / SBCs:** ARM Cortex-M, STM32, NXP/LPC, ESP32/ESP-IDF, Raspberry Pi, PIC, AVR, 8051
- **Embedded Linux & HMI Development:** Raspberry Pi OS/Linux, Python, PySide6/Qt, touchscreen kiosk applications, GPIO control, camera and barcode-scanner integration, system services, and startup automation
- **Hardware Interfaces:** UART, SPI, I2C, GPIO, ADC/DAC, PWM, DMA, USB CDC, memory-mapped I/O
- **Debugging & Bring-Up Tools:** JTAG/SWD debugging, oscilloscopes, logic analyzers, serial debugging, firmware trace analysis, CAN analyzer
- **Control & Electromechanical Systems:** ROS 2/micro-ROS, PWM and relay/contactor-based motor control, ADC feedback, sensor integration, power management, safety interlocks, watchdogs, fault handling, and real-time control logic
- **Signal Processing:** Embedded DSP, FFT, IIR/FIR filters, Butterworth filtering, MATLAB-based signal analysis, CMSIS-DSP
- **Connectivity & Protocols:** TCP/IP, lwIP, MQTT/MQTTs, Wi-Fi, BLE, Modbus RTU, RS-232/485, CAN, BACnet
- **Programming Languages:** C, C++, Python, C#, R
- **Development & Design Tools:** ESP-IDF, Keil MDK, MPLAB, Visual Studio, Git, Altium Designer, Eagle, KiCad, OpenSCAD, Azure IoT Hub
- **Memory & Boot Systems:** EEPROM, NOR/NAND Flash, SPI Flash, NVS, bootloaders, OTA-safe firmware updates
- **Security:** TLS, AES, ChaCha20-Poly1305, MQTT over TLS, post-quantum cryptography (Kyber, Dilithium)

WORK EXPERIENCE

Graduate Research Assistant

January 2023 – August 2026

South Dakota State University, Brookings, South Dakota

- Architected and developed Embedded C firmware on ESP32/ESP32-S3 using FreeRTOS for real-time sensor acquisition, device drivers, Modbus RTU communication, and secure MQTT/TLS telemetry over Wi-Fi.
- Designed and integrated MCU drivers for sensors and peripherals, enabling reliable data acquisition from soil moisture, temperature, humidity, nitrate, and chloride sensing systems.
- Performed hardware bring-up, peripheral integration, and low-level debugging for sensor-based embedded systems using SPI, I2C, UART, RS-485, GPIO, and ADC interfaces.
- Supported PCB design, schematic review, and board-level integration using KiCad for custom IoT sensing hardware.
- Engineered secure telemetry pipelines using AES, MQTT over TLS, Kyber, Dilithium, and ChaCha20-Poly1305, with backend validation and visualization using C#/.NET tools.
- Designed and validated embedded systems using structured testing, debugging, and verification workflows to ensure reliable operation in real-world greenhouse and hydroponic environments.
- Designed and 3D-printed custom OpenSCAD prototypes with integrated sensor mounts, leak-resistant fluid containment, drainage ports, and hose interfaces for leachate collection.
- Developed a Raspberry Pi 5-based automated conveyor control system using Python and PySide6, integrating a touchscreen kiosk interface, proximity sensor, barcode scanner, camera, and relay/contactor motor controls; implemented direction control, timed image capture, fault detection, safety timeouts, and controlled system shutdown.

Assistant Professor

August 2017 - December 2022

Cosmos College of Management and Technology, Kathmandu, Nepal

- Taught embedded systems, microcontroller programming, and real-time firmware development.
- Led embedded R&D projects involving MCU firmware, sensor integration, and hardware–software systems.

CTO / Technical Consultant

April 2017 - December 2022

Samadhan Engineering Pvt. Ltd., Kathmandu, Nepal

- Led embedded systems development from concept to deployment, including requirements review, firmware architecture, hardware integration, testing, documentation, and production validation.
- Developed real-time firmware using PWM, timers, ADC feedback, RTOS scheduling, sensor interfaces, and communication protocols on microcontroller platforms.
- Developed CAN bus driver code for an access management system involving motorized door control, and used a CAN analyzer to inspect, verify, and debug CAN message traffic during system integration.
- Designed and reviewed PCB layouts using Altium Designer for embedded sensing and control systems, supporting board bring-up, testing, and hardware–firmware debugging.

- Developed BLE-based telemetry for a structural health monitoring prototype to transmit damage-index data, later migrating the system to Wi-Fi for improved deployment reliability.
- Designed embedded firmware architectures for field-deployed systems with robust error handling, communication reliability, and maintainable driver-level code.
- Established Git-based version control and YAML-based CI/CD workflows using Docker images to generate repeatable firmware builds and manage release artifacts for embedded systems.

Research Scientist

March 2016 - March 2017

KAIST - Measurement Robotics & Diagnostic AI Lab, Daejeon, South Korea

- Developed embedded DSP firmware with lwIP TCP/IP networking for wireless structural monitoring systems.
- Implemented high-speed signal acquisition and wireless communication pipelines on ARM-based embedded platforms.
- Supported sensor electronics integration, signal validation, and embedded system testing for real-time structural monitoring applications.

Research Assistant

August 2013 - February 2016

Jeonbuk National University, Jeonju, South Korea

- Developed firmware for a dual-core ARM Cortex-M4/M0 LPC4370 system for ultrasonic signal generation, synchronized data acquisition, and real-time DSP processing.
- Configured the LPC4370's built-in USB peripheral as a virtual COM port to transfer ultrasonic waveform data to a laptop for verification before migrating wireless transmission to an external Bluegiga Wi-Fi modem.
- Designed and supported PCB layouts using Eagle for embedded ultrasonic sensing hardware and assisted with board bring-up and hardware debugging.
- Developed Qt-based GUI tools to control a laser excitation system, acquire ultrasonic wave data from an embedded sensing device, and visualize/analyze received waveforms for structural monitoring experiments.
- Implemented FFT, IIR, and FIR filtering algorithms for embedded structural monitoring applications.
- Supported hardware–software integration, signal validation, and debugging for real-time sensing systems.

Firmware Engineer

March 2011 - August 2013

Real Time Solutions Pvt. Ltd., Lalitpur, Nepal

- Developed embedded C/C++ firmware for ARM, PIC, and 8051 microcontrollers, including device drivers, communication stacks, and RTOS-based task scheduling.
- Implemented Protothreads-based cooperative multitasking on 8051 and PIC microcontrollers and developed custom bare-metal schedulers for resource-constrained embedded systems.
- Implemented communication protocols including UART, SPI, Modbus RTU, and cellular telemetry using AT commands.
- Developed firmware for industrial and field-deployed systems including panel meters, MPPT solar battery chargers, queue management systems, and flood early-warning devices.
- Debugged hardware–firmware interactions using oscilloscopes, logic analyzers, and serial debugging tools to improve system reliability.

EDUCATION		
• Ph.D. , Agricultural & Biosystems Engineering - South Dakota State University, USA		Aug 2026
• Graduate Certificate , Data Science - South Dakota State University, USA		Aug 2025
• M.E. , Mechatronics Engineering - Jeonbuk National University, South Korea		Dec 2016
• B.E. , Electronics & Communication - Cosmos College of Management and Technology, Nepal		Dec 2010

SELECTED PROJECTS

- **Multichannel Ultrasonic Structural Health Monitoring Device (STM32 Cortex-M4 + ESP32):** Developed real-time STM32 firmware for ultrasonic excitation, synchronized high-speed acquisition, and FFT/IIR processing, with ESP32-based wireless communication and display functions. Wrote an I2C OLED driver and performed hardware bring-up and debugging.
- **Distributed IoT Monitoring System with Post-Quantum Security (ESP32-S3, FreeRTOS):** Built a multi-node telemetry system for real-time nitrate and soil-moisture monitoring. Integrated sensor drivers, FreeRTOS task coordination, MQTT/TLS, Kyber, Dilithium, ChaCha20-Poly1305, backend verification, and performance testing.
- **MPPT-Based Solar Battery Charger (PIC MCU):** Implemented embedded firmware for an MPPT solar charger using ADC-based voltage and current feedback with PWM control to optimize energy conversion under changing solar conditions.
- **Automated Plant-Stem Conveyor Imaging System (Raspberry Pi 5, Python/PySide6):** Built a Linux-based imaging and machine-control system integrating a touchscreen HMI, barcode scanner, proximity sensor, camera, and relay/contactors controls. Implemented stop-settle-capture-restart sequencing, bidirectional operation, watchdogs, fault recovery, and safe shutdown.
- **Panel Meter with Energy Accumulation and Modbus RTU (NXP Cortex-M0):** Developed firmware for an LCD-based panel meter with push-button controls, precise energy accumulation, and a fourth-order Butterworth IIR filter for signal conditioning. Implemented a custom Modbus RTU slave interface for communication with external data loggers.

PUBLICATIONS (SELECTED)	
• Structural Health Monitoring (SAGE): Wireless ultrasonic device for aircraft structures (2018).	
• Smart Agricultural Technology (ELSEVIER): Distributed IoT + PQC for nitrate/soil-moisture monitoring in corn production (2026).	